Lecture 2 Planned Notes

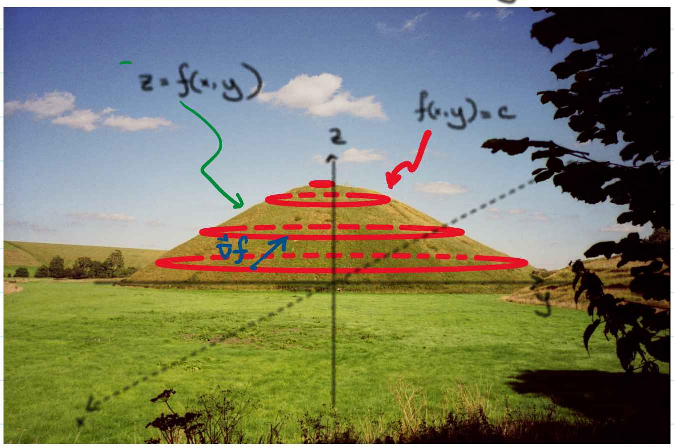
Sunday, May 11, 2025 11:44 PM

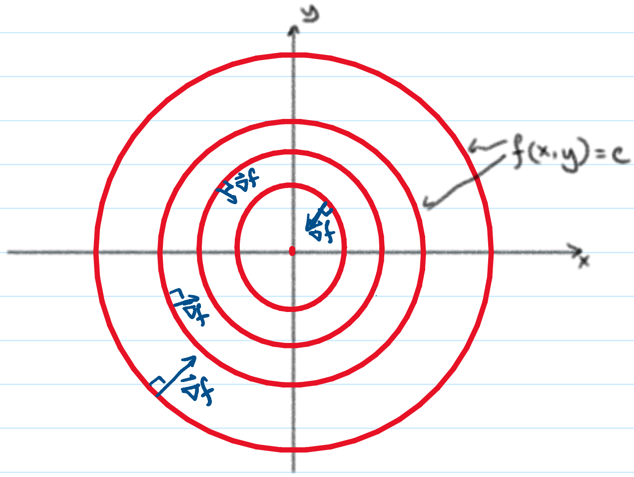
Review:

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

For a scalar function $f(x,y,z)$. $\vec{\nabla} f$ is a vector field that is

- the direction of maximum increase
- normal to level surfaces $f(x,y,z) = \text{constant}$





Directional derivative: $D_{\vec{u}} f = \vec{u} \cdot \vec{\nabla} f$ where $|\vec{u}| = 1$.